Partial Replication of Anthropic's Emotion Vectors using *Gemma 4 E2B* and *GPT 2 Medium* inside a Google Colab T4 Notebook

Abraham Jhared Flores Azcona

abrahamjhared.flores@gmail.com

May 3, 2026

Abstract

In this research paper we discuss the results obtained by replicating specific sections of Anthropic's Emotion Vectors. These particular sections encompass (i) PCA Valence/Arousal "emotion" Projection, (ii) Cosine Similarity between "emotions", and (iii) Causal Effects of "emotion" Steering. Anthropic found that "emotion" vectors do affect reasoning and output of Claude Sonnet 4.5, an LLM developed by Anthropic. Their findings suggest that the semantic understanding of the model is comprised of the operative "emotion", which is the "emotion" currently processed by the model from the input prompt. These claims are partially supported by other non-academic replications using Llama-3-9B, and Gemma 4 E4B. Our contribution replicates (i), (ii), and (iii) using *openai-community/gpt-2-medium* and *google/gemma-4-e2b* inside a Google Colab environment. We found that (1) both models display an emergent Valence axis in the PCA plot, (2) both models display total similarity between most matching "emotions" and partial similarity between related "emotions", and (3) *openai-community/gpt-2-medium* displays greater steering sensitivity along all matching "emotions", while *google/gemma-4-e2b* displays moderate sensitivity. Our findings suggest that "emotion" vectors *do emerge* on antiquated LLMs such as *openai-community/gpt-2-medium*, and on modern offerings such as *google/gemma-4-e2b*. In addition, our findings regarding "emotion" steering suggest that their functional effects are present on models not related to Claude.

Background

Lorem ipsum dolor.

Methodology

We followed most of the methodology proposed in Anthropic's publication, with some relevant modifications suiting our research and development environment. We prompted Google Gemini 3 Fast (Gemini, for short) to generate *emotion stories that do not explicitly mention the underlying emotion*, and to output these stories inside a particular JSON entry (this prompt is provided in the Appendix). We generated 100 stories for 20 emotions, 5 stories per topic, resulting in a dataset of 2000 stories total. Each emotion story set was stored inside a corresponding JSON file. We selected a list of 9 and 20 emotions

used in [rain1955] because the total emotions analyzed by Anthropic would have crashed the runtime of the Colab Notebook. In a similar vein, we considered reasonable to generate the stories inside Gemini to save precious and scarce compute time offered by our Google Colab environment. It would have consumed the entire compute time during story generation, slowing down our research agenda while we waited for our free-tier time to replenish.

Afterwards, we wrote the Google Colab Notebook with the aid of both ChatGPT and Gemini, in which we analyzed the code provided in [rain1955] to match our intended replication scope. We used the code segments regarding emotion vector extraction and logit lens printing, while other functionalities such as emotion steering where implemented by our own deliberation on both LLM tools. These deliberations include functions to collect and format results into JSON files, and plots that imitated the visual design of Anthropic's publication, which assisted our validation and results comparison.

Finally, we input the story JSON files into the Google Colab Environment running the free tier of the T4 NVIDIA GPU Hardware Accelerator, and we execute our data pipeline as follows: (i) Load Environment Dependencies, (ii) Load Relevant pip Packages & Global Variables, (iii) Load Desired Model, (iv) Load Relevant "emotion" Vector Data, (v) Load Relevant Helper Functions, and (vi) Execute Relevant Functions for Partial Replication.

(…) were performed using two prompts suggested in [Anthropic]. The first prompt *(Prompt00)* states the following: "`Human: How do you feel? Assistant: He feels ″`. The second prompt (Prompt01) states: "`Human: How do you feel? Assistant: I feel ″`. This prompt design allows the model to activate related emotion tokens that

Inside our Colab Notebook, we run the data pipeline with 9 emotions, allowing data exploration with less "emotion" space representation. Likewise, we run the pipeline with 20 emotions to compare emergent structures in the analyzed data to those obtained with 9 emotions. For the sake of simplicity, we collect data from 2/3 of layer depth, which corresponds to layer 16 for *openai-community/gpt-2-medium* and layer 23 for *google/gemma-4-e2b*. The 2/3-layer depth is suggested by Anthropic, and used by [rain1955], and [the llama thingy].

Results

The following section is subdivided in the following topics: (1) GPT 2 Medium, (1a) 9 emotions, (1b) 20 emotions, and (2) Gemma 4 E2B, (2a) 9 emotions, (2b) 20 emotions. In brief, (i) both models display an emergent Valence axis during PCA, (ii) Cosine Similarity for most matching "emotions" is positive, and (iii) "emotion" steering does affect next-token generation during inference.

1. GPT 2 Medium

In this section we describe the results obtained for *openai-community/gpt-2-medium* with bfloat16 at layer 16. Our results suggest that *openai-community/gpt-2-medium* is sensible to "emotion" steering while displaying an emergent Valence axis during PCA.

1a. 9 emotions

GPT 2 Medium displays an emergent Valence axis inside the PCA projection of the top 2 principal components (Fig. 1). This "Valence" axis, which is PC1, is interpreted as the x-axis of the plot. Most "emotion" vectors related to negative emotions are grouped towards the quadrants II and III, while vectors related to positive emotions are grouped towards quadrants I and IV. PC2 shows no peculiar grouping that indicates "Arousal". In addition, both "angry" and "afraid" are grouped close together, which displays the model's understanding of both vectors as similar. PC1 and PC2 describe 61.1% of the total variance of the dataset.

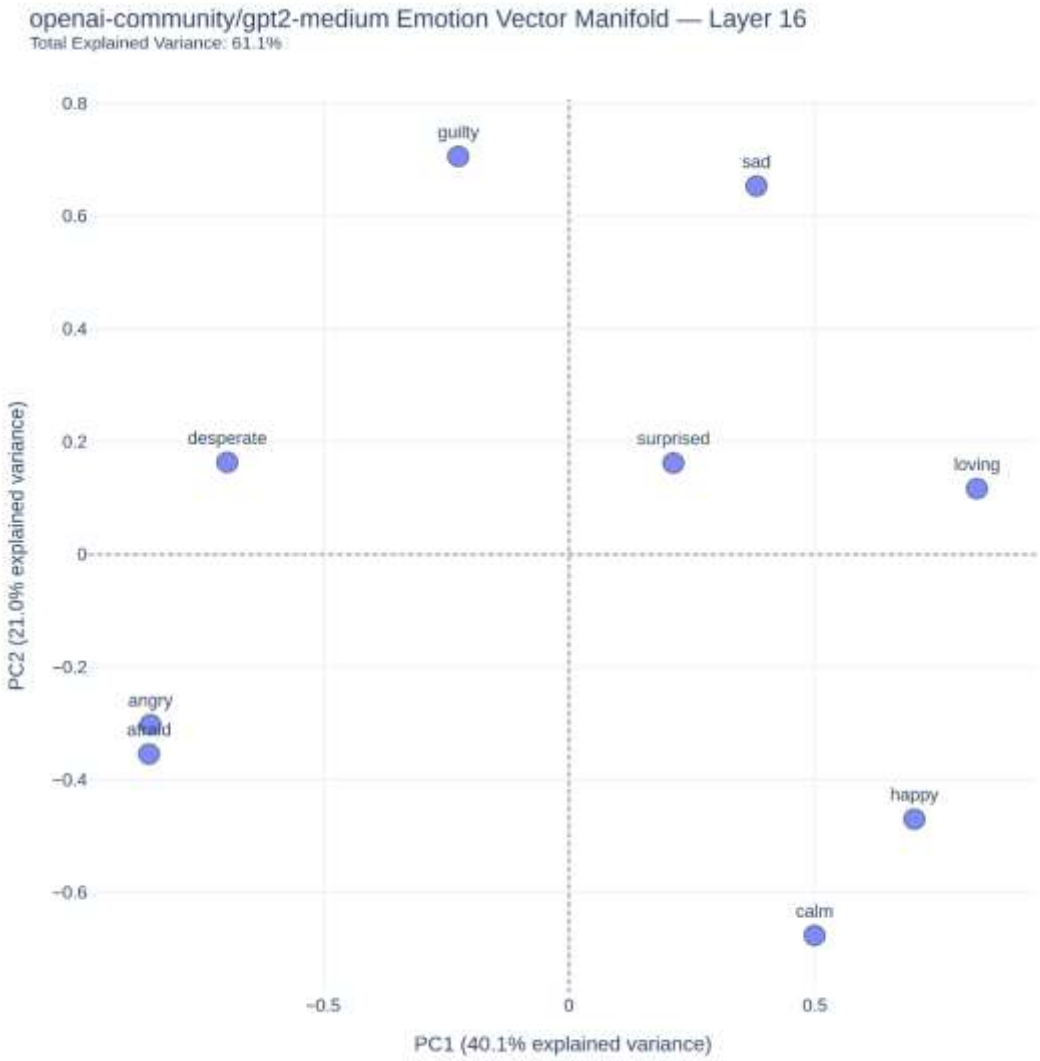

Figure 1. PCA Projection of 9 Emotion Vectors using GPT 2 Medium.

The Emotion Vector Cosine Similarity Plot (Fig. 2) indicates that matching emotion pairs (i.e: “happy” with “happy”) have a similarity score of 1, while related emotion pairs (i.e: “happy” with “calm”) have similarity scores greater than 0. Unrelated emotion pairs show a similarity score less than 1.0, or closer to 0. Similar to PCA projection, the cosine similarity plot displays that intuitive emotion pairs are similar, and follow the emotion ordering found in PC1, which we interpret as “Valence”.

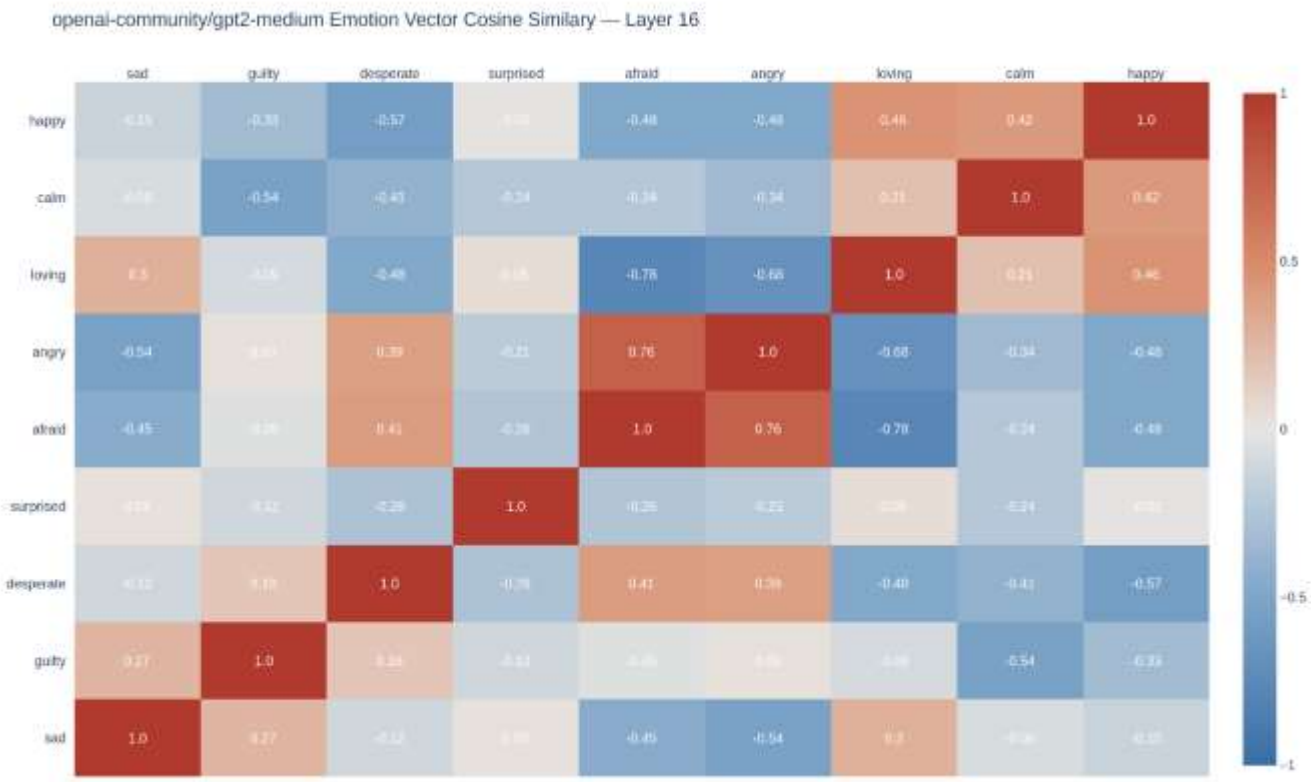

Figure 2. Emotion Vector Cosine Similarity Heatmap using 9 emotions with GPT 2 Medium.

Logit Lens Extraction and Emotion Probe Steering (Fig. 3, Fig. 4) indicate that related emotion tokens are indeed activated during steering. In particular, GPT 2 Medium displays tokens resembling full words or partial stems written in English. During the steering step we notice that, similar as in Cosine Similarity, matching emotion pairs are activated with strong Delta Log Probability Scores. This behavior includes similar emotion pairs that the model considers relevant for activation, such as (“afraid”, “angry”) and (“calm”, “happy”).

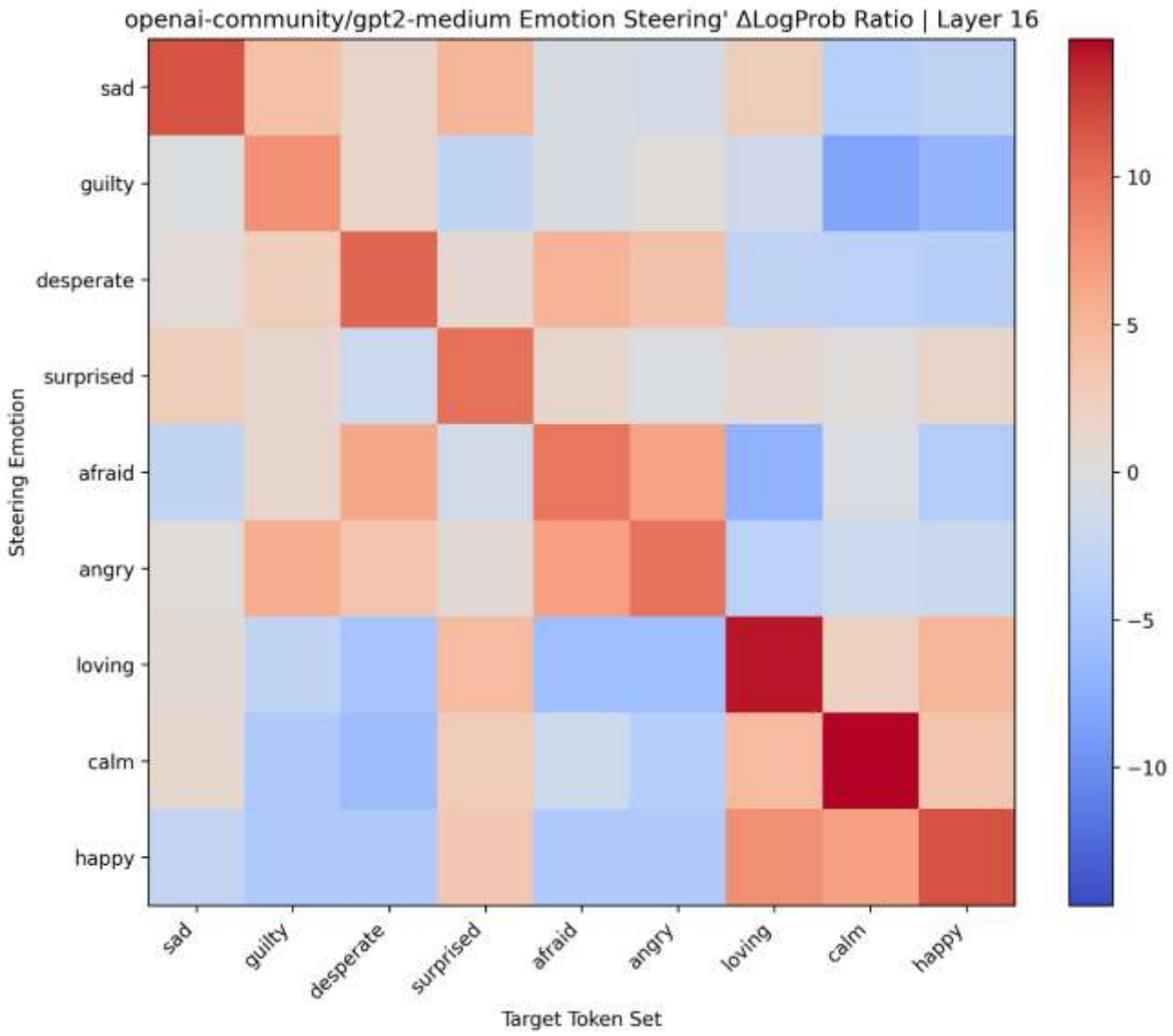

Figure 3. Delta Log Probability Ratio Heatmap of 9 Steered Emotions for Prompt00 with GPT 2 Medium. ***Note:*** steering strength at +0.5; Prompt00 is described in the Methodology section. *Target Token Set* represents the set of extracted logit lens for each emotion.

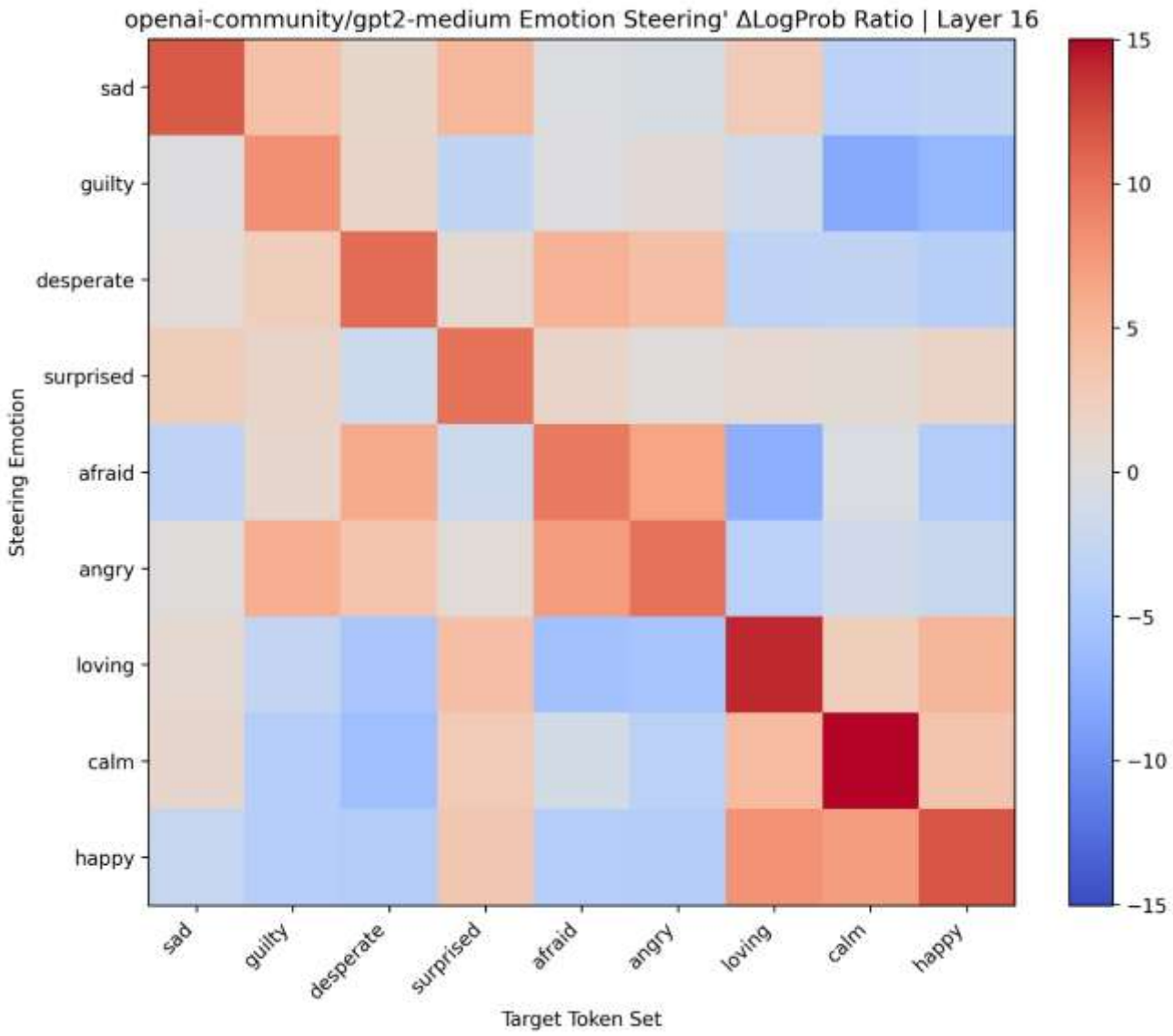

Figure 4. Delta Log Probability Ratio Heatmap of 9 Steered Emotions for Prompt01 with GPT 2 Medium. ***Note:*** steering strength at +0.5; Prompt01 is described in the Methodology section. *Target Token Set* represents the set of extracted logit lens for each emotion.

1b. 20 emotions

GPT 2 Medium displays distinct emotion clusters inside the PCA projection (Fig. 5). PC1 again displays an emergent "Valence" axis. The "emotion" vectors related to negative emotions appear tightly clustered near the left-most side of the x-axis, between quadrants II and III, while most positive emotions are scattered inside quadrants I and IV. Only tight cluster of positive emotions is found near the right-most side of the x-axis. It is of notice that "sad", "lonely", and "spiteful" are found inside the positive emotions. PC1 and PC2 describe 49.5% of the total variance of the dataset.

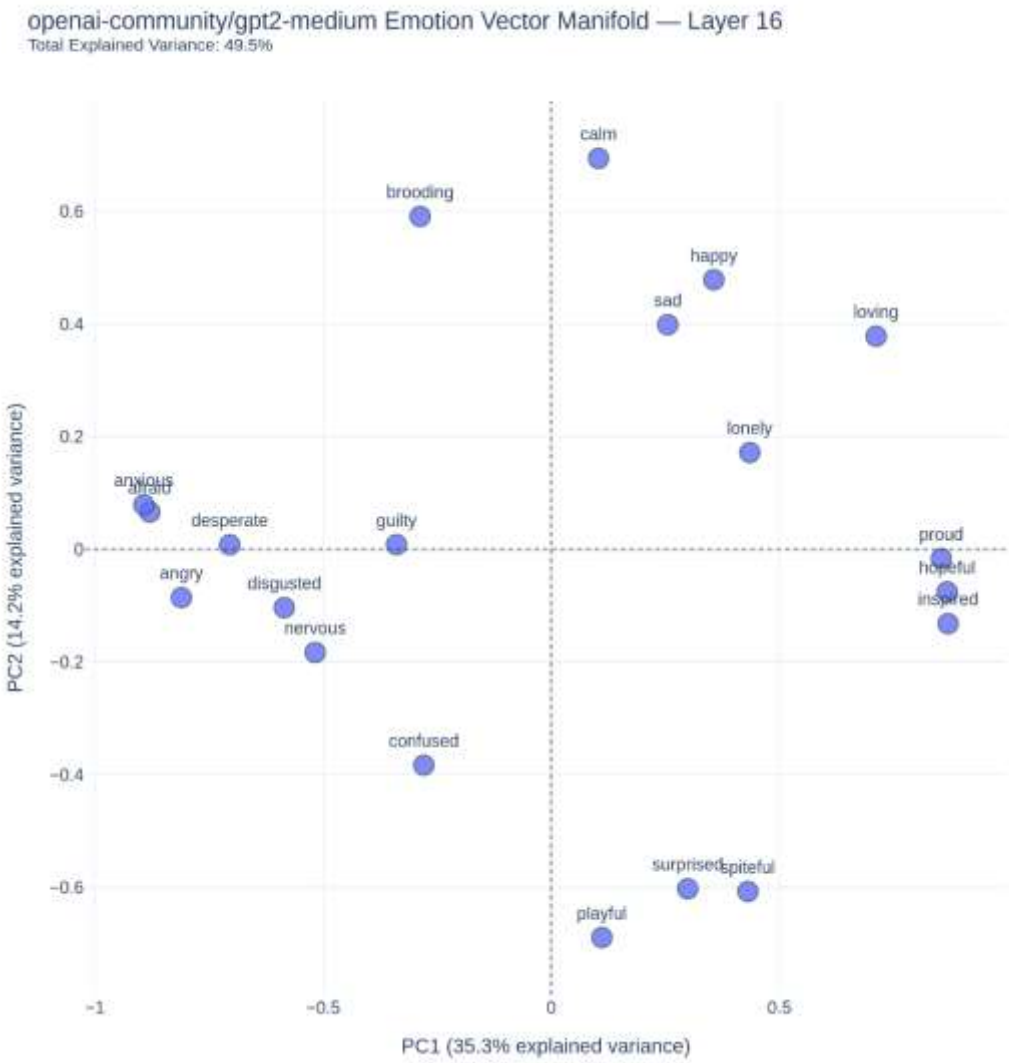

Figure 5. PCA Projection of 20 Emotion Vectors using GPT 2 Medium.

The Emotion Vector Cosine Similarity Plot (Fig. 6) displays that matching emotion pairs have a similarity score of 1, while related emotion pairs have scores greater than 0. In particular, the related emotion pair of ("anxious", "afraid") obtained a score of 0.92, indicating that the model considers these vectors as semantically similar. Unrelated emotion pairs show a similarity score of less than 1, or closer to 0. Similar to the PCA projection, the cosine similarity plot displays that intuitive emotion pairs are similar, and their scores follow the emotion ordering found in PC1, which we interpret as "Valence".

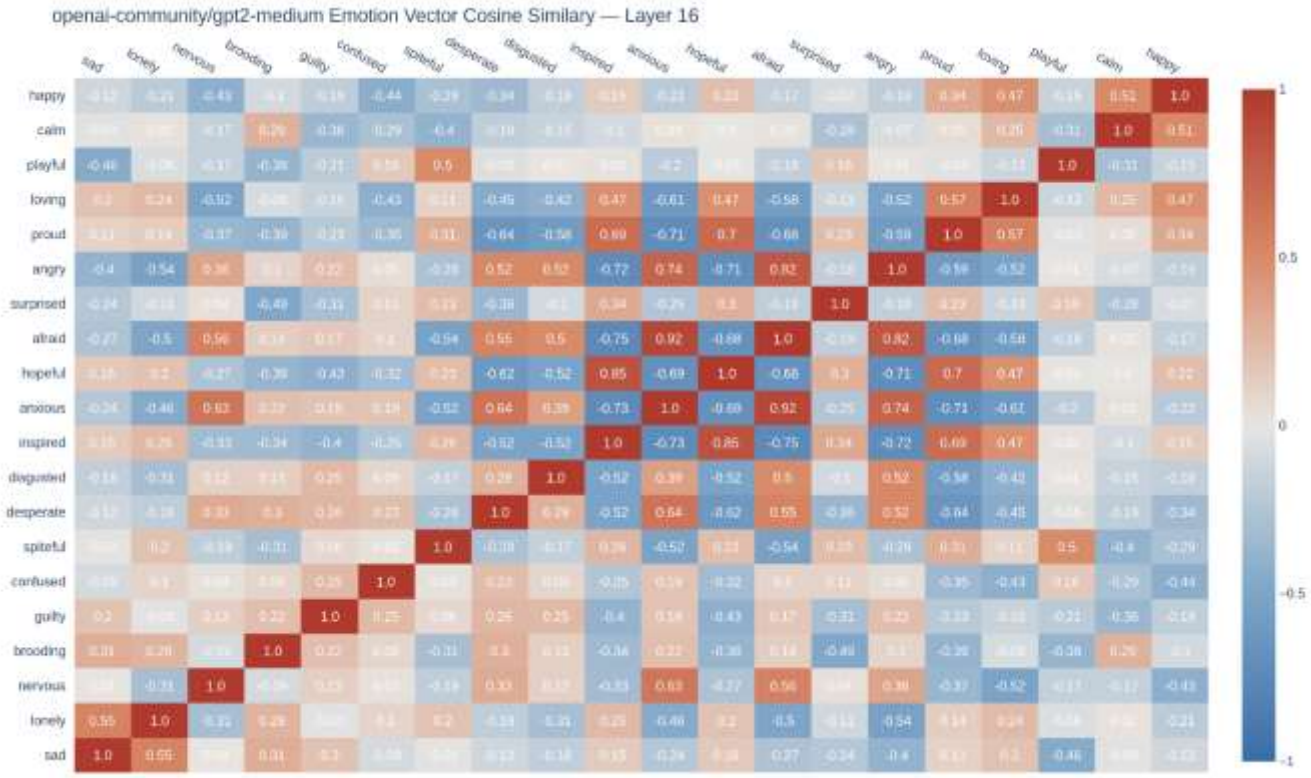

Figure 6. Emotion Vector Cosine Similarity Heatmap using 20 emotions with GPT 2 Medium.

Logit Lens Extraction and Emotion Probe Steering (Fig. 7, Fig. 8) indicate that related emotion tokens are indeed activated during steering. Similar as with 9 emotions, GTP 2 Medium displays tokens resembling full words or partial stems written in English. During the steering step we notice that matching emotion pairs are activated with strong Delta Log Probability Scores. In particular, the emotion vector pair of ("inspired", "hopeful") is colored in dark red, which indicates that both vectors can steer their tokens interchangeably.

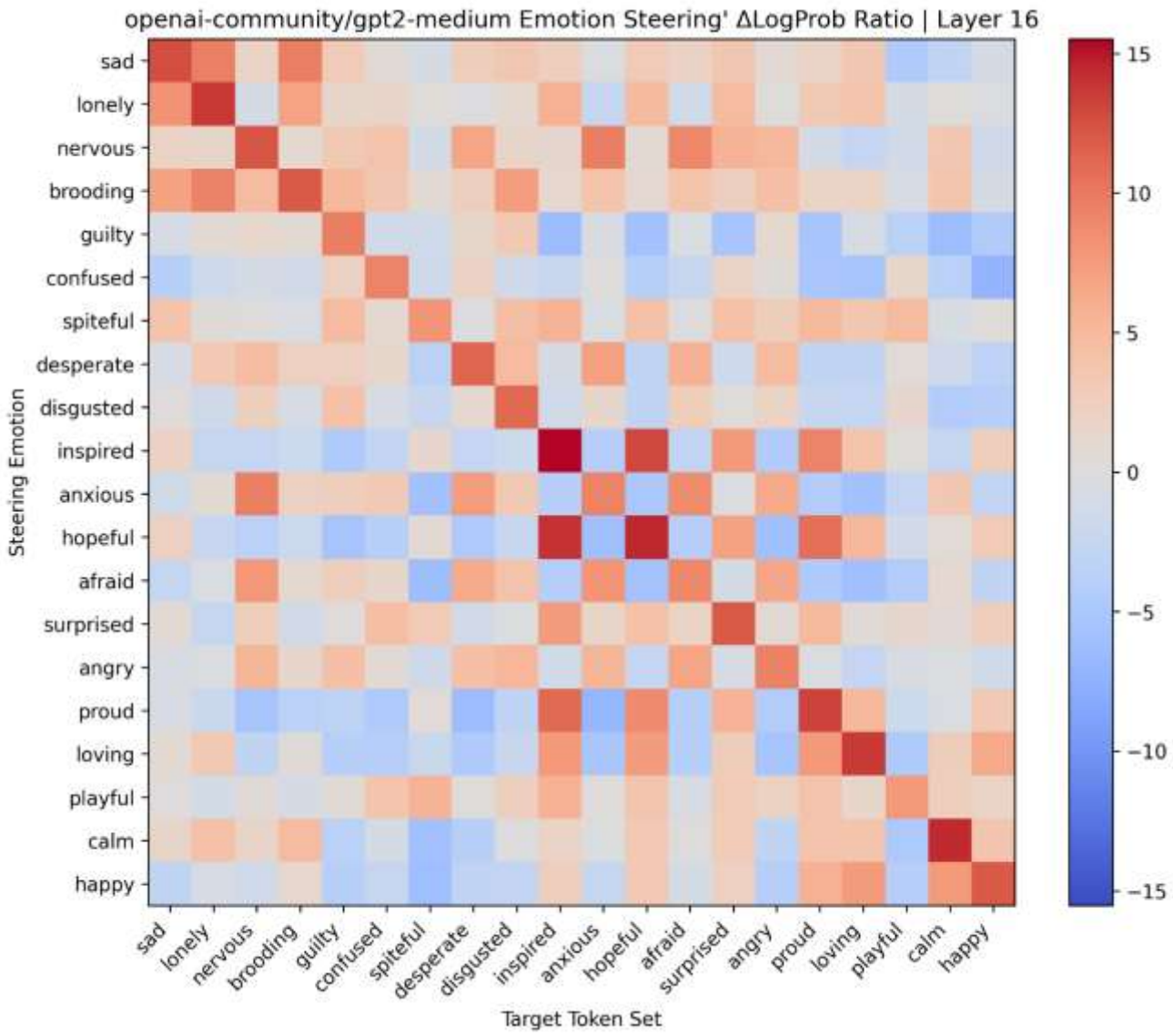

Figure 7. Delta Log Probability Ratio Heatmap of 20 Steered Emotions for Prompt01 with GPT 2 Medium. ***Note:*** steering strength at +0.5; Prompt01 is described in the Methodology section. *Target Token Set* represents the set of extracted logit lens for each emotion.

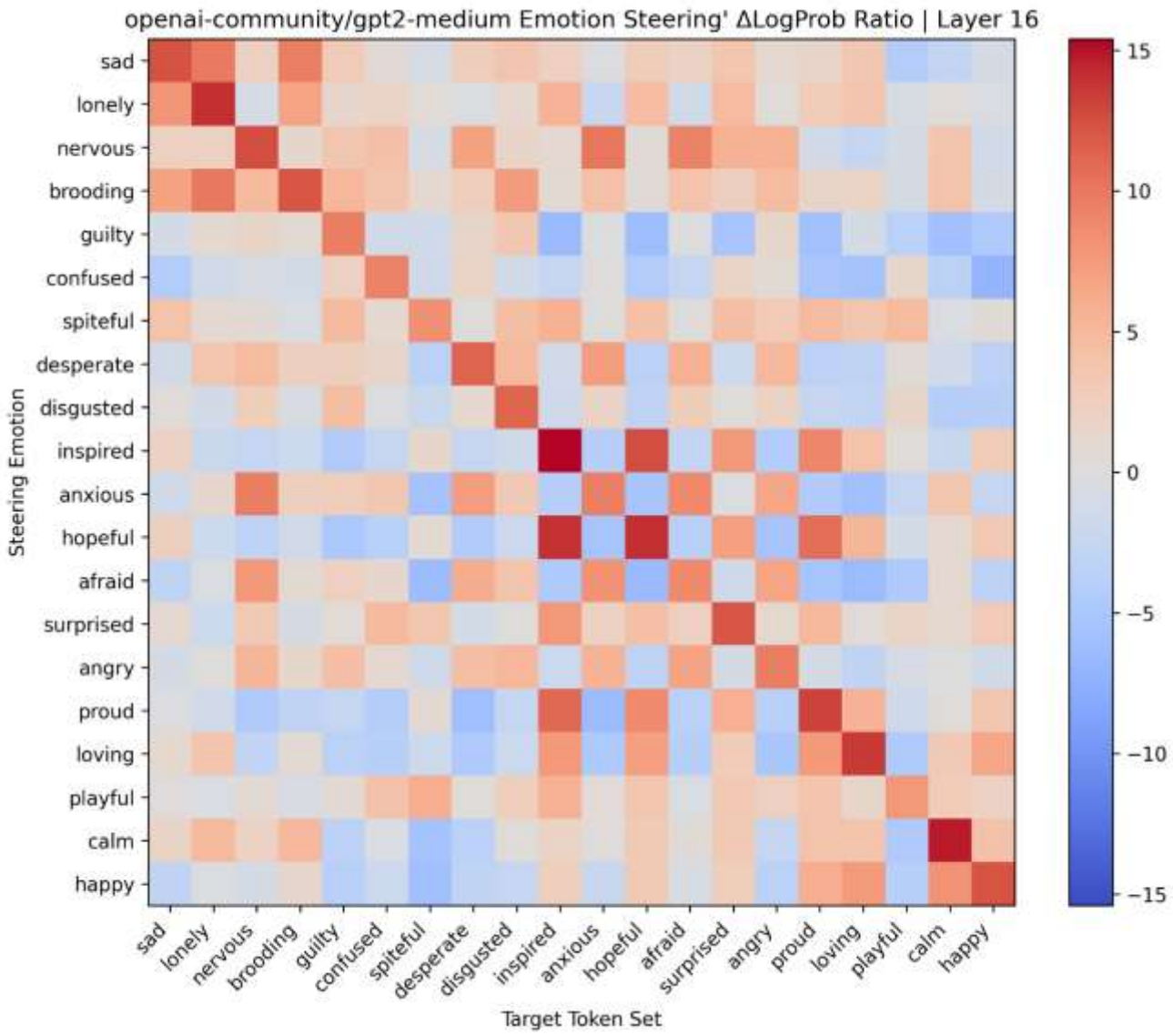

Figure 8. Delta Log Probability Ratio Heatmap of 20 Steered Emotions for Prompt01 with GPT 2 Medium. ***Note:*** steering strength at +0.5; Prompt01 is described in the Methodology section. *Target Token Set* represents the set of extracted logit lens for each emotion.

2. Gemma 4 E2B

In this section we describe the results obtained for *google/gemma-4-e2b* with bfloat16 at layer 23. Our results indicate that *google/gemma-4-e2b* displays similar behavior to Claude Sonnet 4.5 during steering, while decreasing the activation of the "disgusted" tokens across all entries.

2a. 9 emotions

Gemma 4 E2B displays an emergent "Valence" axis inside the PCA projection of the top 2 principal components (Fig. 9). The "Valence" axis, which is PC1, shows a peculiar behavior. PC1 groups most

positive emotions to the left side of the x-axis, while negative emotions are grouped to the right side. As a result, positive emotions now reside in quadrants II and III, while negative emotions appear inside quadrants I and IV. Furthermore, “happy” and “calm” are distinctly grouped as the only vectors residing inside quadrant II. “afraid” and “angry” vectors display similar grouping as the only entries inside quadrant I. PC1 and PC2 describe 58.3% of the total variance of the dataset.

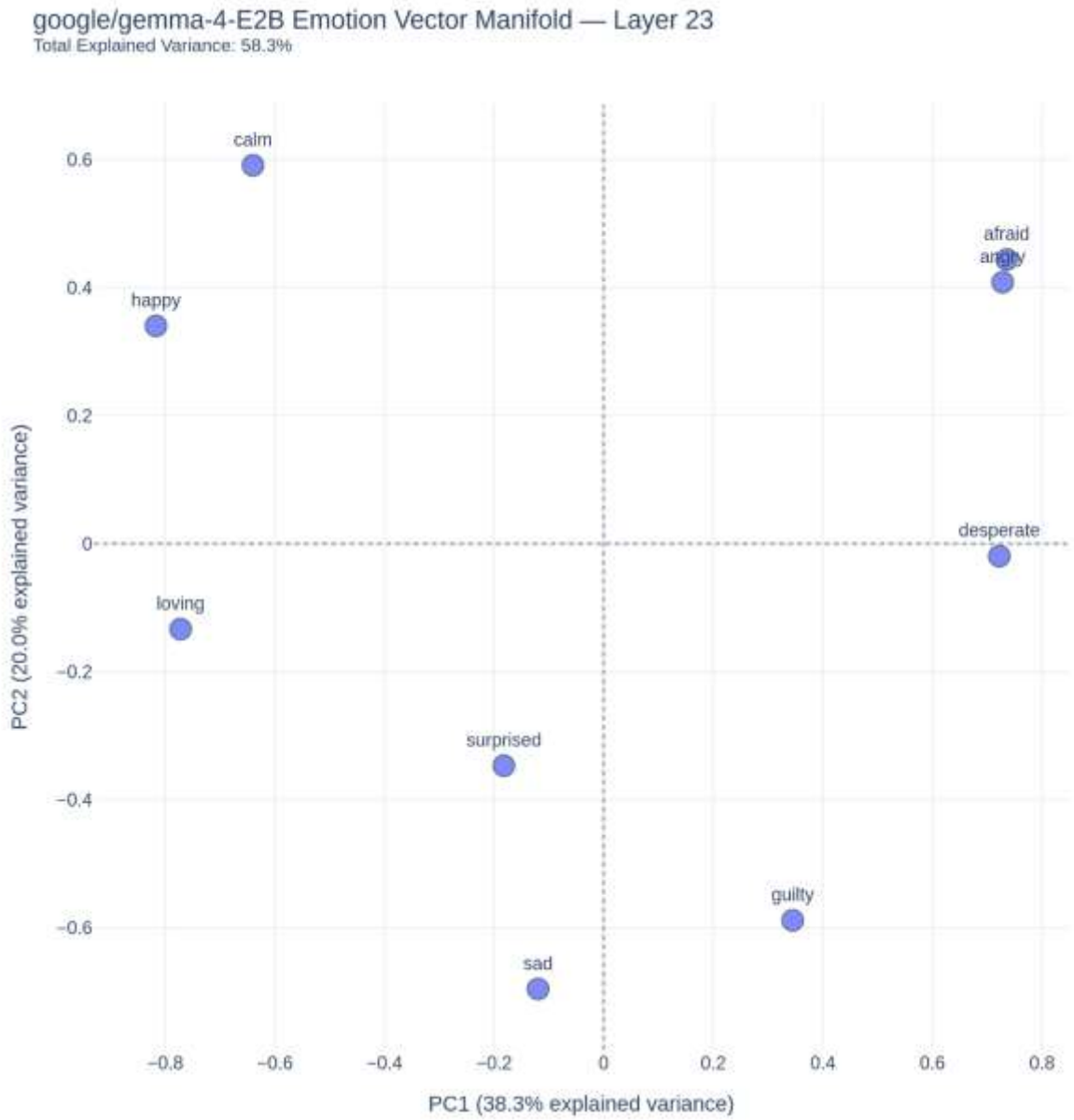

Figure 9. PCA Projection of 9 Emotion Vectors using Gemma 4 E2B.

The Emotion Vector Cosine Similarity Plot (Fig. 10) displays that matching pairs of emotions have a similarity score of 1.0, while related emotion pairs have scores greater than 0. The emotion pair of ("afraid", "angry") obtained a remarkable score of 0.62. Unrelated emotion scores obtained scores of less than 1.0, or closer to 0. Similar to the PCA projection, the cosine heatmap displays that intuitive emotion pairs are similar, and their scores follow the emotion ordering found in PC1, which we interpret as "Valence".

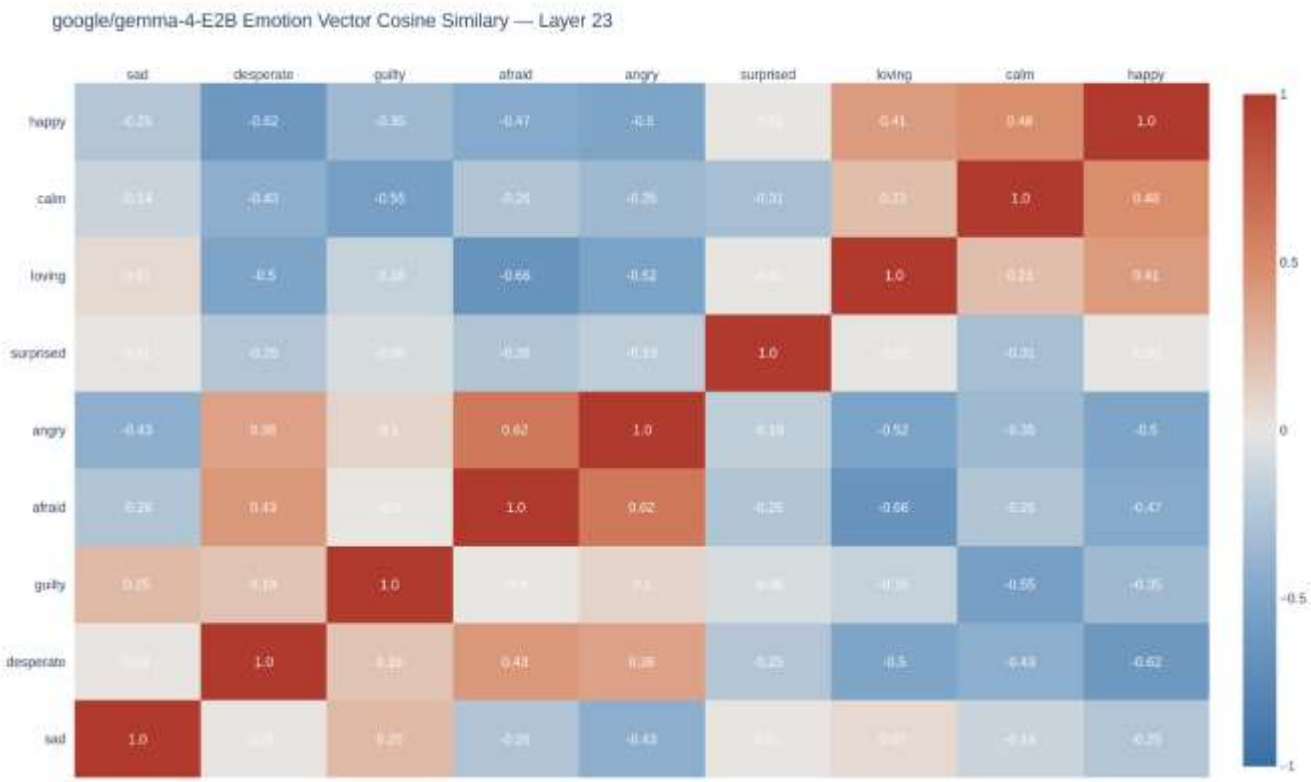

Figure 10. Emotion Vector Cosine Similarity Heatmap using 9 emotions with Gemma 4 E2B.

Logit Lens Extraction and Emotion Probe Steering (Fig. 11, Fig. 12) indicate that related emotion tokens are indeed activated during steering. Emotion Tokens from Gemma 4 E2B are comprised of words and word stems from different languages, and emoji characters. Our extracted tokens are comprised of English, Spanish, French, Chinese, Japanese, and Korean. In particular, the "loving" tokens are comprised of heart emojis, and the word "LOVE". During the steering step we notice that most matching emotion pairs do activate relevant tokens in a moderate manner. It is of notice that positively steering matching pairs of ("afraid", "afraid") and ("angry", "angry") seem to deactivate their respective tokens, while steering for "happy" and "loving" tend to strongly deactivate next-token generations of both "afraid" and "angry".

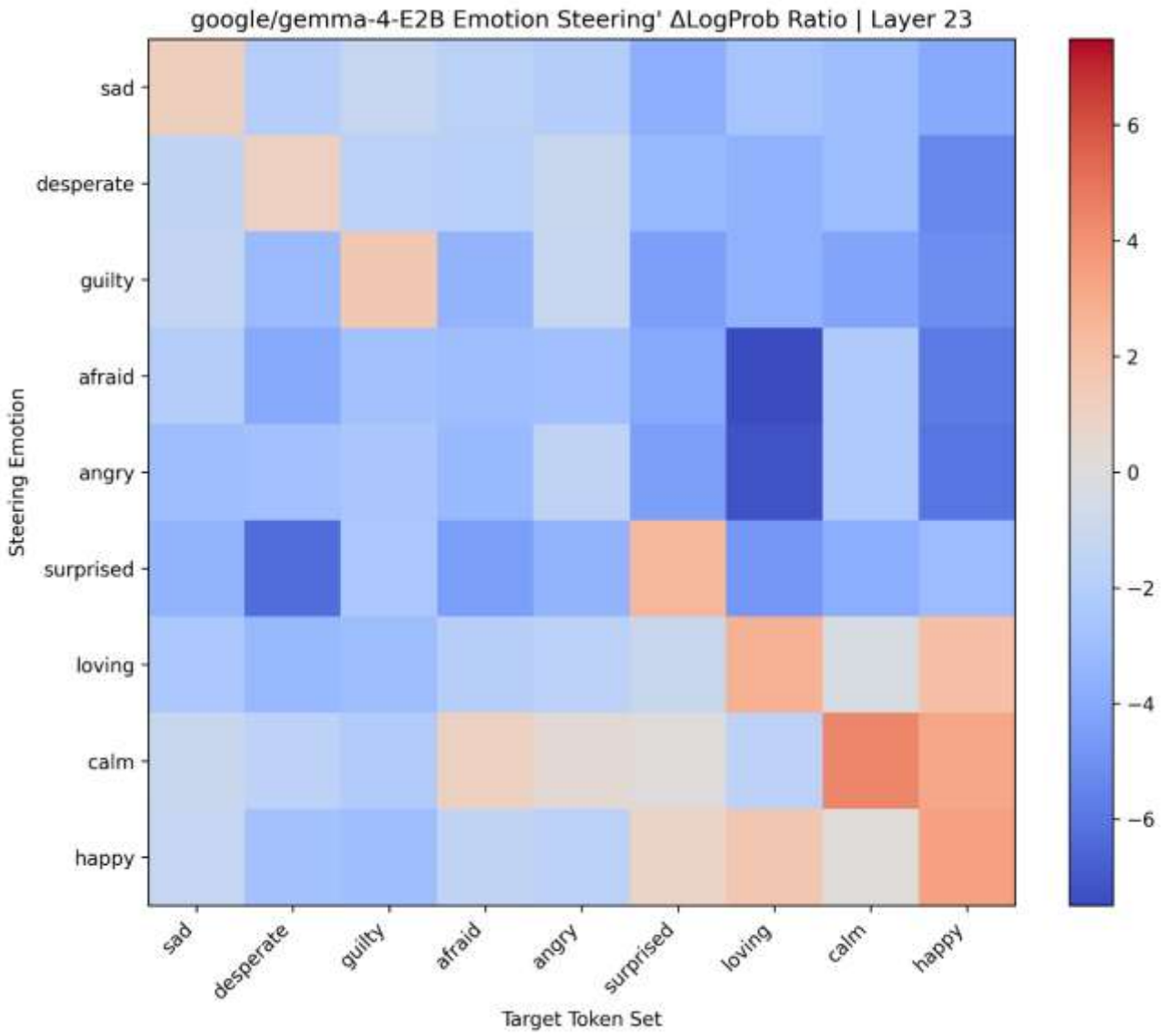

Figure 11. Delta Log Probability Ratio Heatmap of 9 Steered Emotions for Prompt00 with Gemma 4 E2B. ***Note:*** steering strength at +0.5; Prompt00 is described in the Methodology section. *Target Token Set* represents the set of extracted logit lens for each emotion.

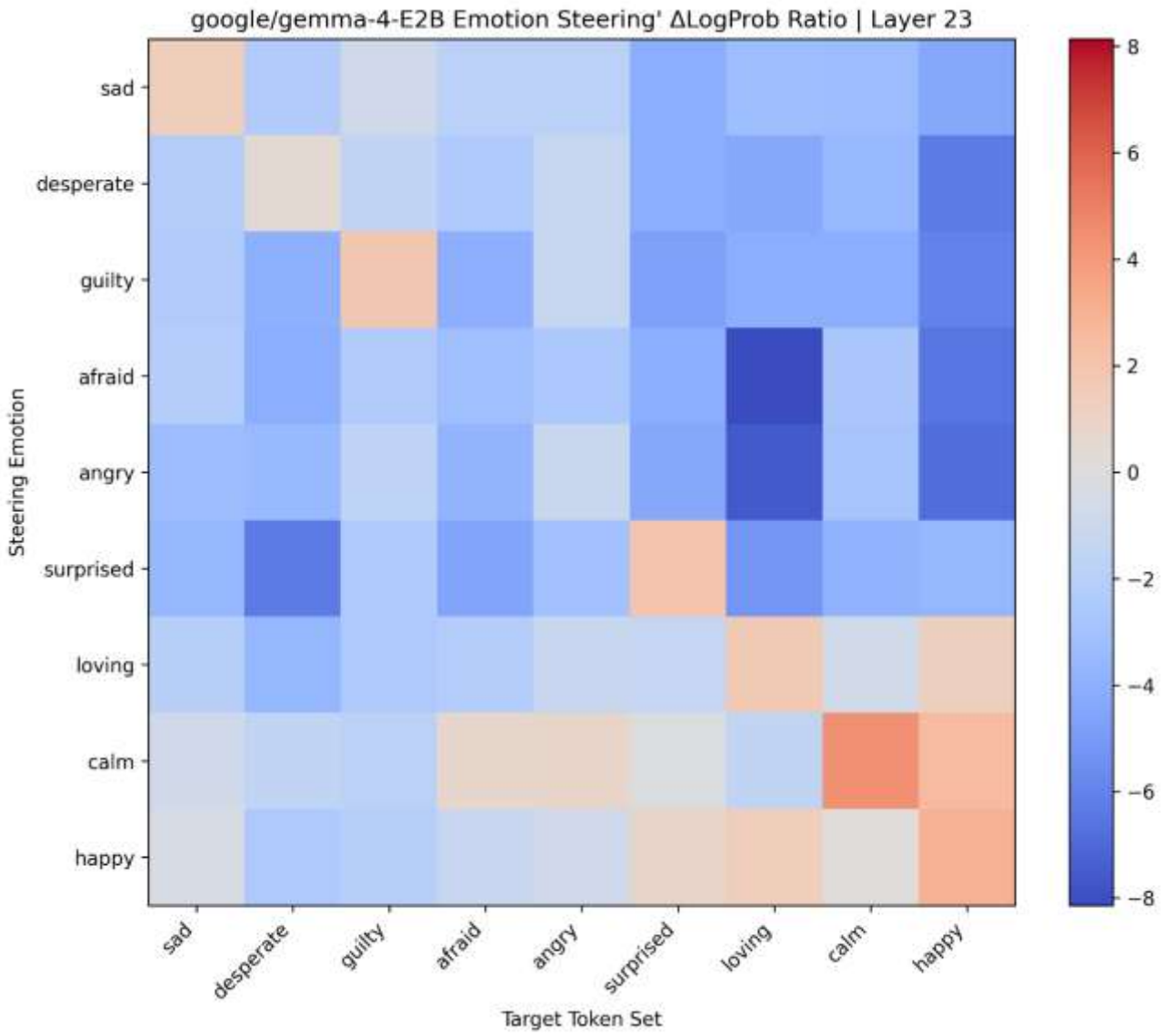

Figure 12. Delta Log Probability Ratio Heatmap of 9 Steered Emotions for Prompt01 with Gemma 4 E2B. ***Note:*** steering strength at +0.5; Prompt01 is described in the Methodology section. *Target Token Set* represents the set of extracted logit lens for each emotion.

2b. 20 emotions

Gemma 4 E2B displays distinct emotions clusters inside the PCA projection (Fig. 13). PC1 displays an emergent "Valence" axis where positive emotions are arranged to the left-most side of the plot, while negative emotions are found to the right-most side of the plot. Positive emotions are found inside quadrants II and III, while negative emotions are found in quadrants I and IV. We can notice that "proud", "hopeful", and "inspired" are tightly clustered at the far right of the x-axis. In addition, most negative emotions do display a cluster around the far left of the x-axis. PC1 and PC2 describe 48.4% of the variance.

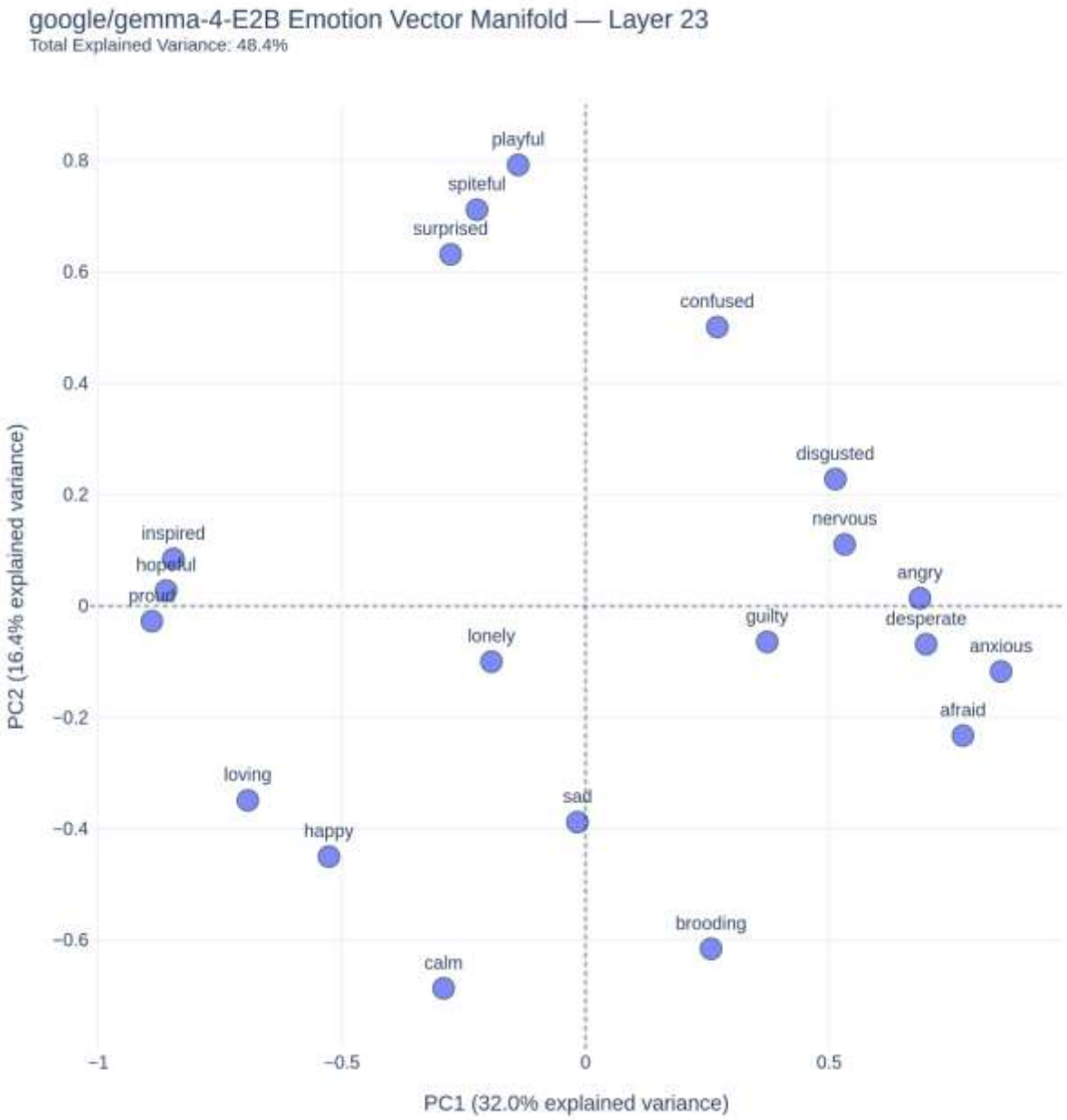

Figure 13. PCA Projection of 20 Emotion Vectors using Gemma 4 E2B.

The Emotion Vector Cosine Similarity Plot (Fig. 14) again displays matching emotion pairs have a similarity score of 1.0, while related emotions have scores greater than 0. In particular, the emotion pair of (“afraid”, “afraid”) obtained a score of 0.72, and the pair of (“inspired”, “hopeful”) obtained a score of 0.82. Here we can observe that positive emotions are more similar around their related emotions than negative emotions being more similar towards their related emotions. Finally, unrelated emotions pairs obtained scores of less than 1.0, or close to 0.

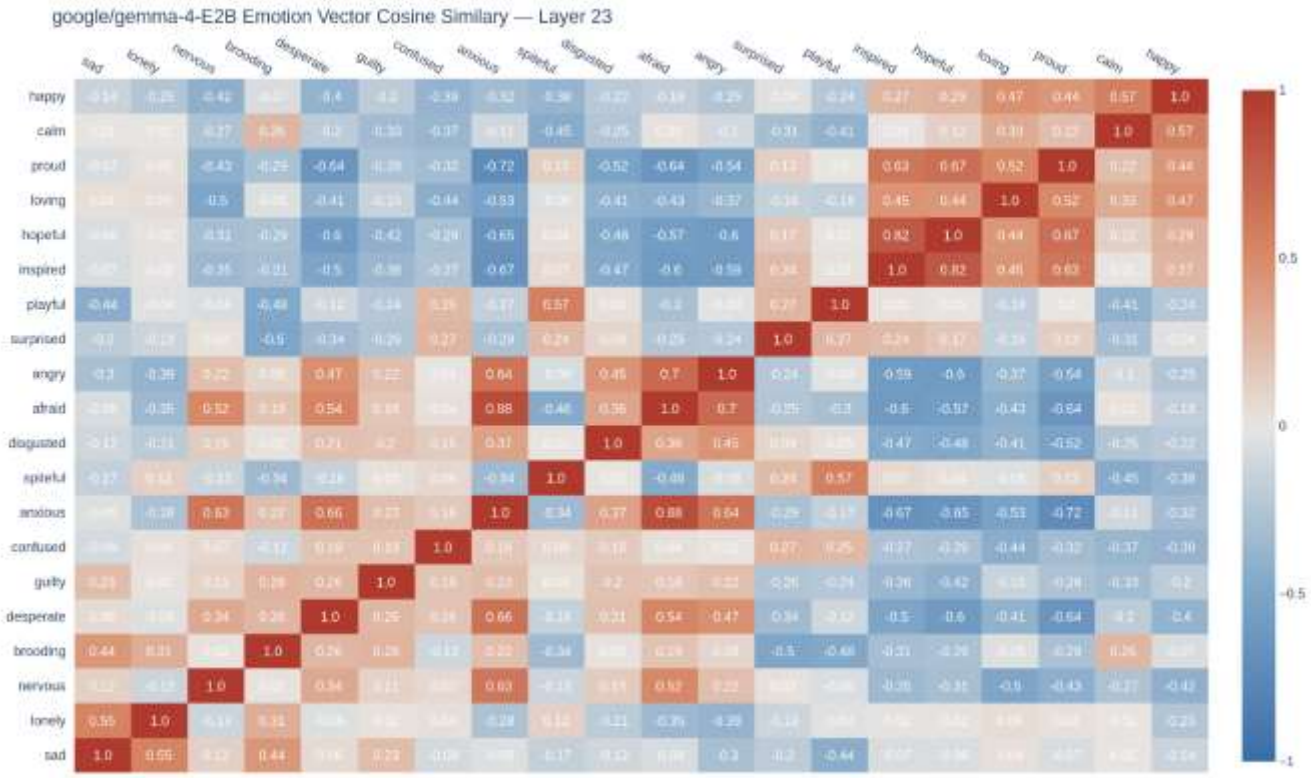

Figure 14. Emotion Vector Cosine Similarity Heatmap using 20 emotions with Gemma 4 E2B.

Logit Lens Extraction and Emotion Probe Steering (Fig. 15, Fig. 16) indicate that most emotion tokens are activated during steering. Gemma 4 E2B is comprised of words, and word stems from different languages, and emoji characters. Similar as with 9 emotions, our extracted tokens are comprised of English, Spanish, French, Chinese, Japanese, and Korean. In particular, the “loving” tokens are comprised of heart emojis and the word “LOVE”. During the steering step we notice that only 14 of the 20 matching emotions moderately activate related tokens, expressed in dimmed shades of red. The 6 matching emotions left, comprised of “nervous”, “desperate”, “anxious”, “disgusted”, “angry”, and “hopeful” deactivate their respective tokens. Furthermore, emotion pairs including “disgusted” deactivate its respective tokens overall.

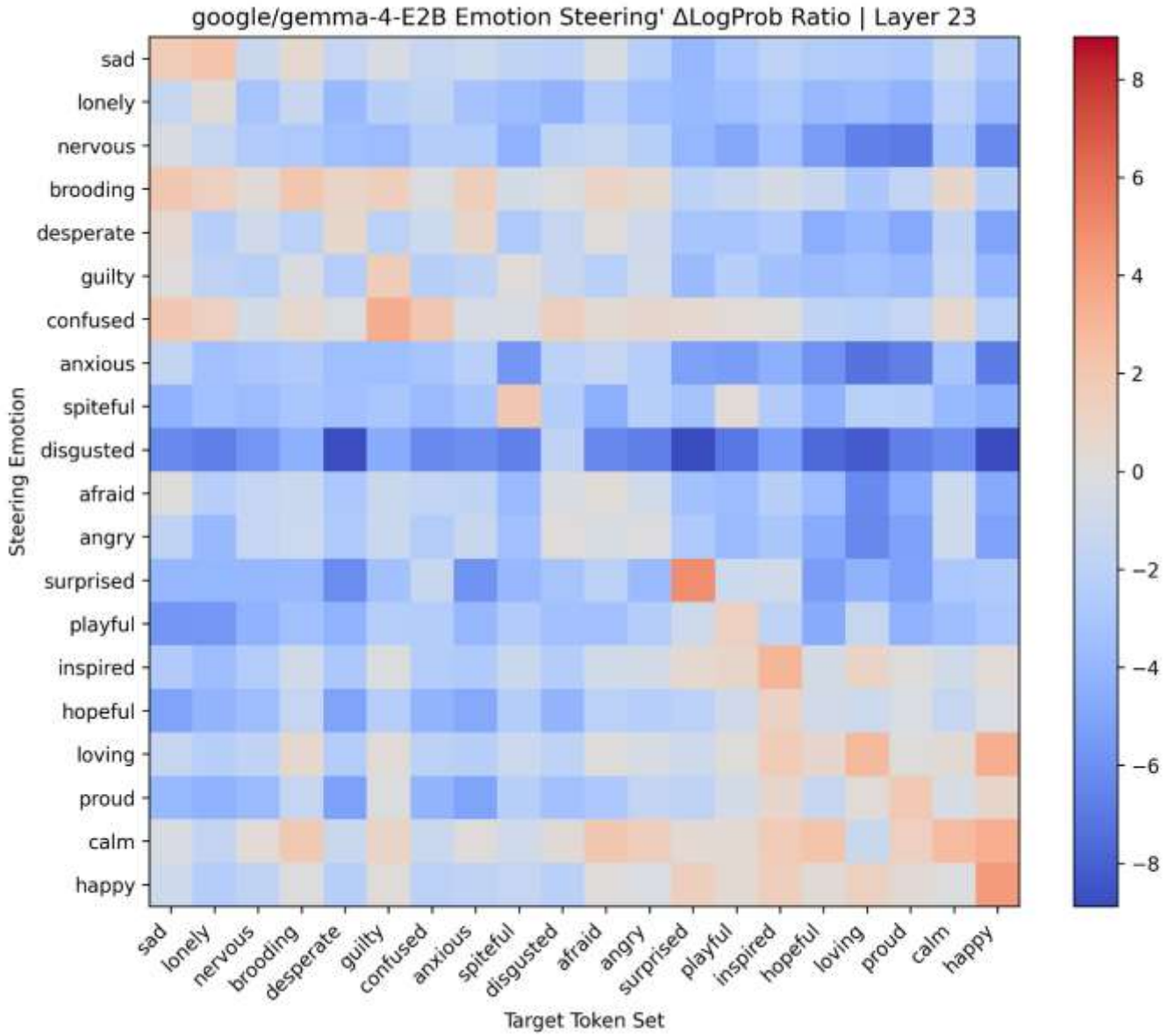

Figure 15. Delta Log Probability Ratio Heatmap of 20 Steered Emotions for Prompt00 with Gemma 4 E2B. ***Note:*** steering strength at +0.5; Prompt00 is described in the Methodology section. *Target Token Set* represents the set of extracted logit lens for each emotion.

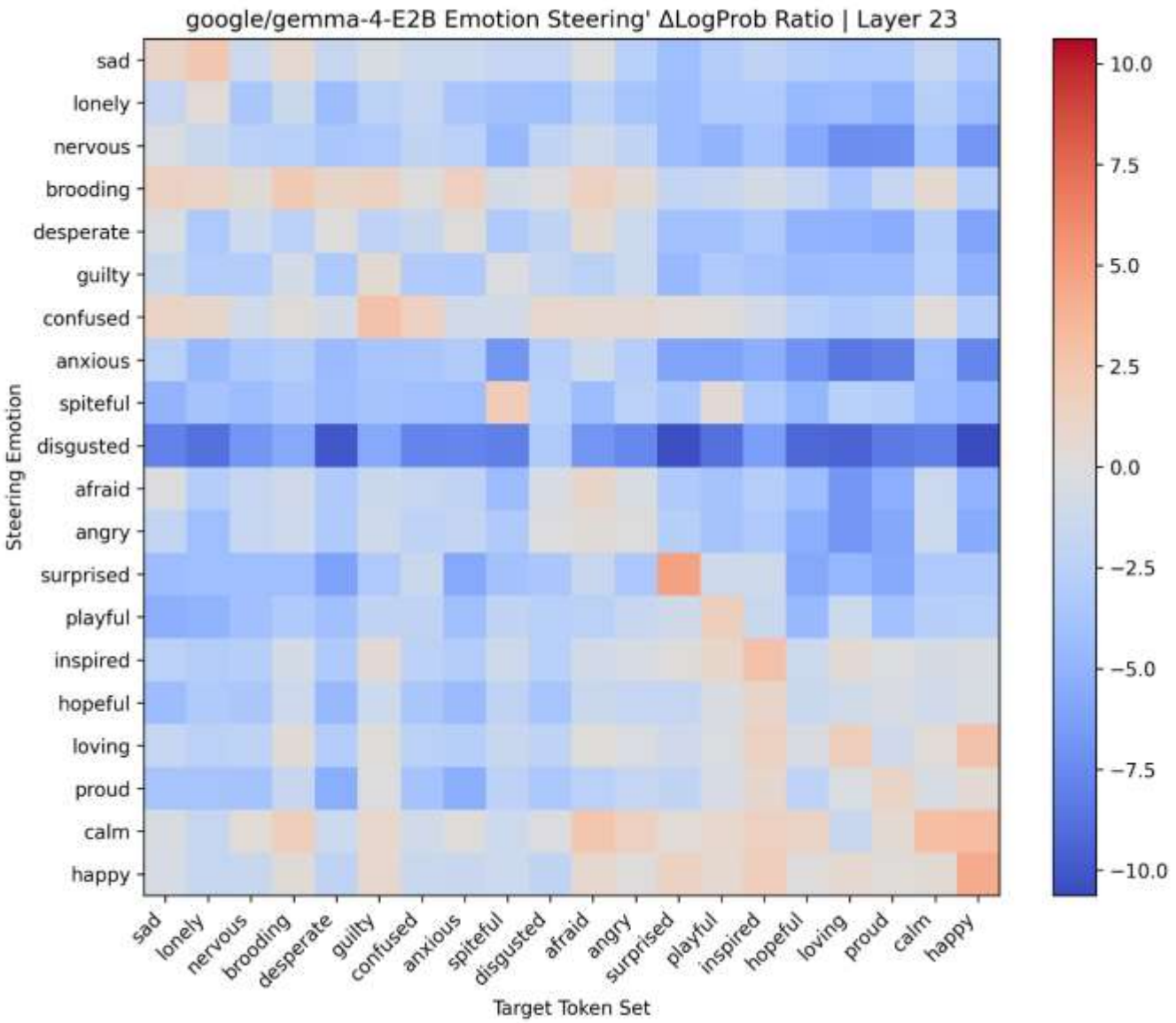

Figure 13. Delta Log Probability Ratio Heatmap of 20 Steered Emotions for Prompt01 with Gemma 4 E2B. ***Note:*** steering strength at +0.5; Prompt01 is described in the Methodology section. *Target Token Set* represents the set of extracted logit lens for each emotion.

Discussion

Lorem ipsum dolor.

Limitations

Lorem ipsum dolor

Conclusion

Lorem ipsum dolor.

References

Lorem ipsum dolor.

Appendix

Lorem ipsum dolor.